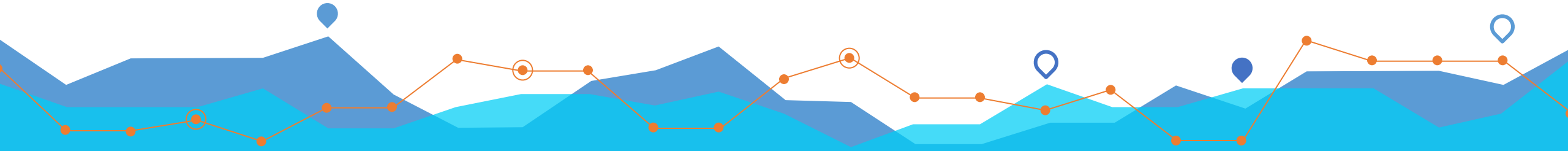


## Lecturers:

Rafael Gonçalves (102534)

André Butuc (103530)



# TapID: Rapid Touch Interaction in Virtual Reality using Wearable Sensing

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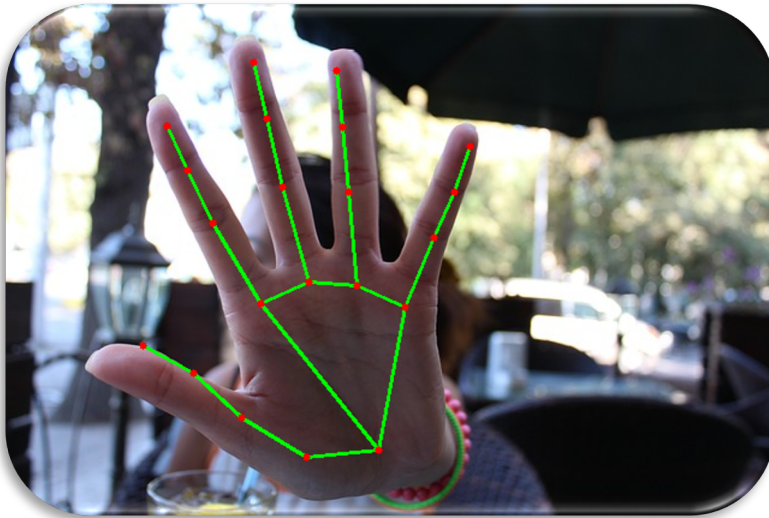
**Date of Conference:** 27 March-1 April 2021

**Number of pages:** 10 pages

# HANDS: OUR BEST TOOL

## IN AUGMENTED REALITY

- Hand tracking is well-suited to accompany the work on physical objects, for example in maintenance and repair.



## IN VIRTUAL REALITY

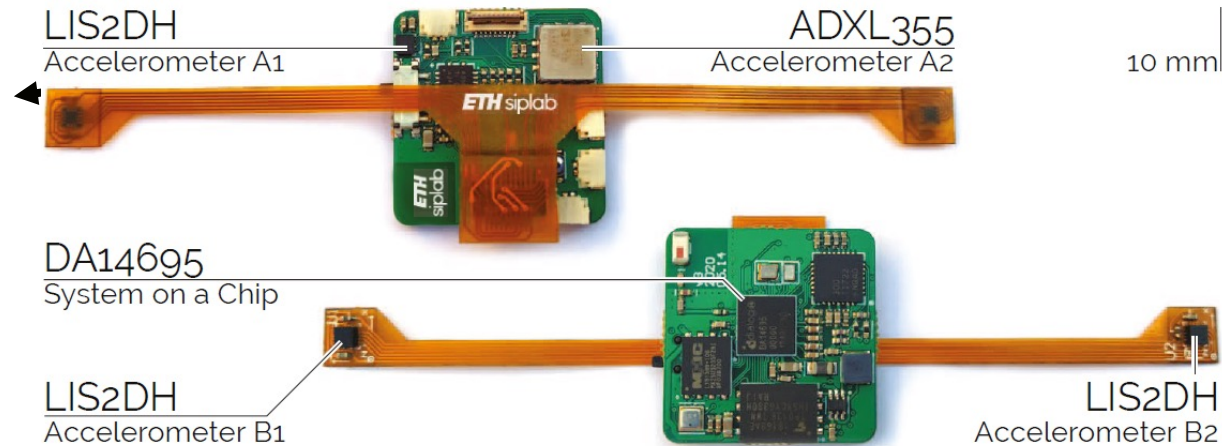
- Free-hand interaction
  - ❑ exceeds quick interactions
  - ❑ has lack of boundary awareness

## WHY NOT MOVING ALL DIRECT INTERACTION IN VR TO PASSIVE SURFACES?

- **LESS FATIGUE FOR PROLONGED INTERACTION**

# TapID AS A SOLUTION

HEAD-MOUNTED SYSTEMS  
LEAD TO **INACCURACY** IN  
DEPTH SENSING WHILE  
TRACKING HANDS



- ❖ Touch interaction on surfaces provides users with an opportunity to **rest their arms** between interactions.
- ❖ TapID is a wrist-worn device that **detects taps** on surfaces and identifies the tapping finger, which, combined with tracked hand poses, triggers input in VR.

# STUDY CHALLENGE

HOW TO PROMOTE **PROLONGED INTERACTIONS**  
WHERE THE INPUT IS

- **QUICKER**
- **MORE RELIABLE**
- **NON-FATIGUING**

?





# WHY THIS PAPER

**Multimodal Interfaces**

**Physical Gadget**

**Appealing Abstract**



# PROJECT CONTRIBUTIONS

## First

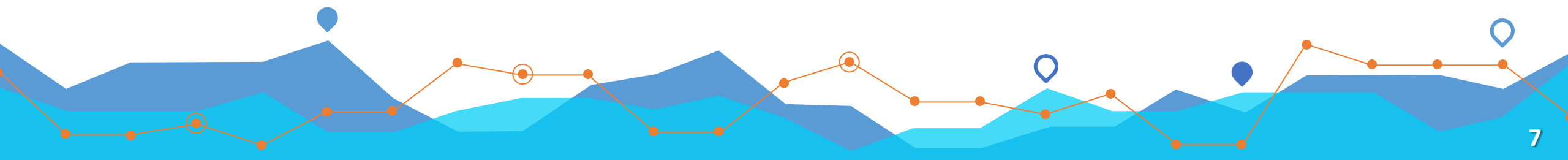
Introduced a wrist-based sensing principle to detect surface touches and to identify the finger that triggered the tap event.

## Second

Demonstrated a prototype implementation (TapID) that processes and classifies input events in real time using a machine learning pipeline.

## Third

Technical evaluation with 18 participants

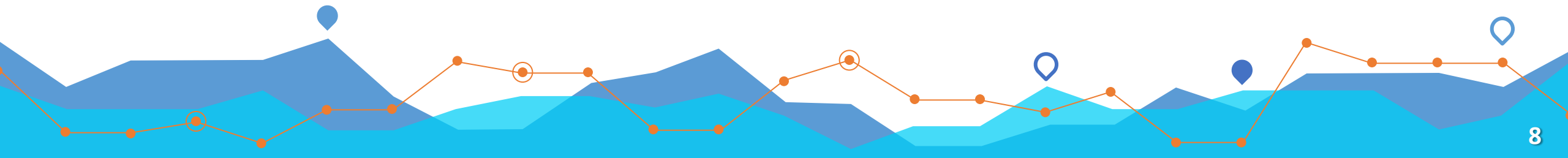
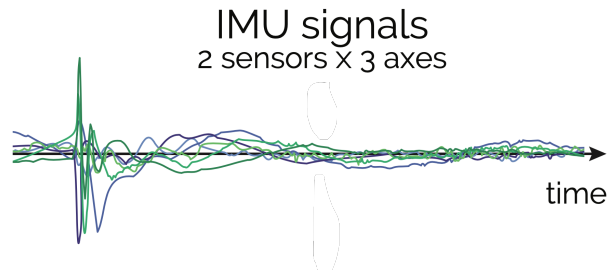


# PHASES OF TOUCH PROCESSING

Tap Event Detection

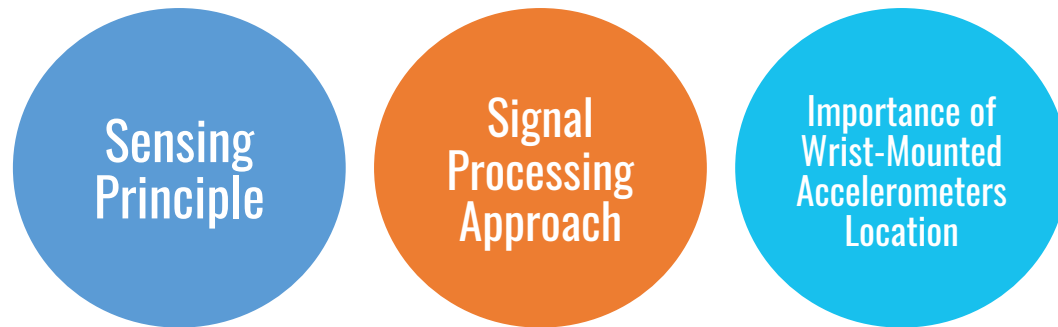


Internal Tap Event Representation



# PERFORMANCE STUDY

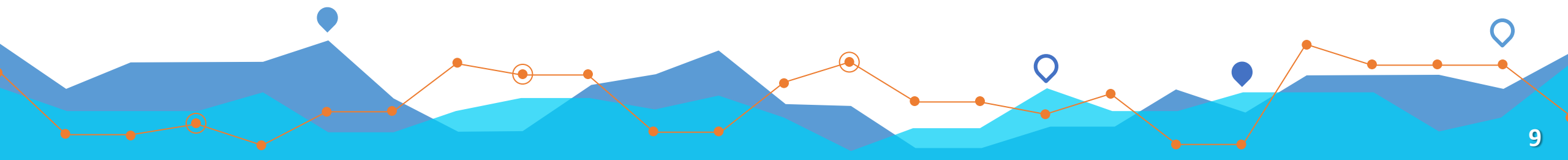
## GOALS



## PARTS

Accuracy of Tap Detection

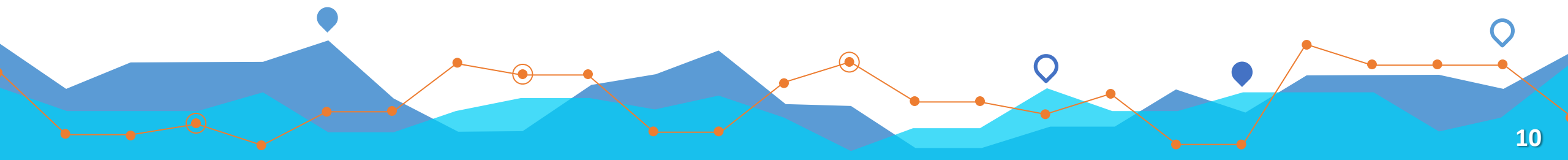
Accuracy of Finger Identification Network



# PERFORMANCE STUDY

## Participants

|                          | Part 1  | Part 2  |
|--------------------------|---------|---------|
| Total Participants       | 18      | 4       |
| Male Participants        | 14      | 3       |
| Female Participants      | 4       | 1       |
| Age's Range (years)      | 19-57   | -       |
| Age's Mean (years)       | 28.4    | 25.3    |
| Wrist Circumference (mm) | 146-185 | 158-176 |



# PERFORMANCE STUDY

## Procedure

**Brief Introduction  
of Tap ID**



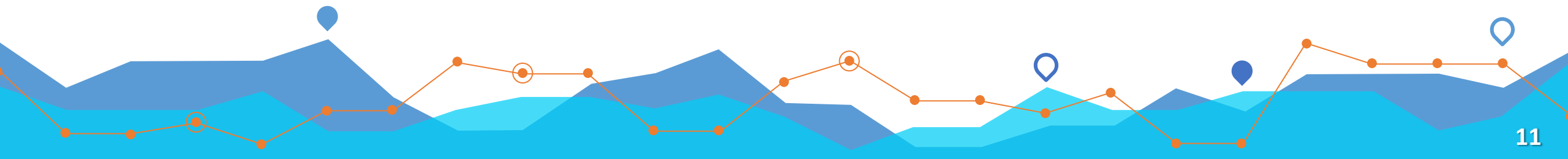
**Gathering the  
Participants Data**



**Instructing  
Participants**



**Finger Tap  
Rounds**





# EVALUATING TAP EVENT DETECTION

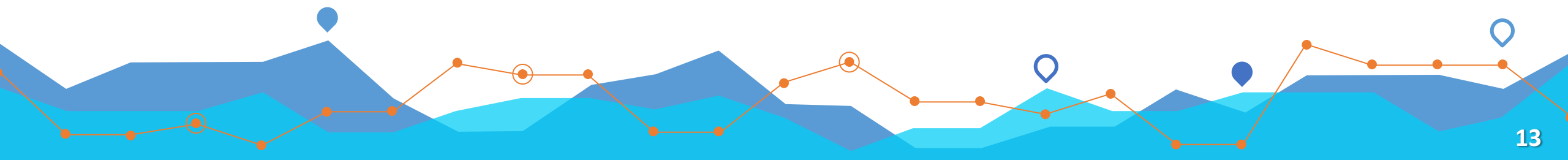
- A **surface acoustic signal** was recorded using a stethoscope that was covered and taped to the tabletop.
- During the evaluation, participants **kept quiet** to avoid introducing artifacts into the recorded signal.
- This sub-evaluation resulted in a total of 3600 tap events.



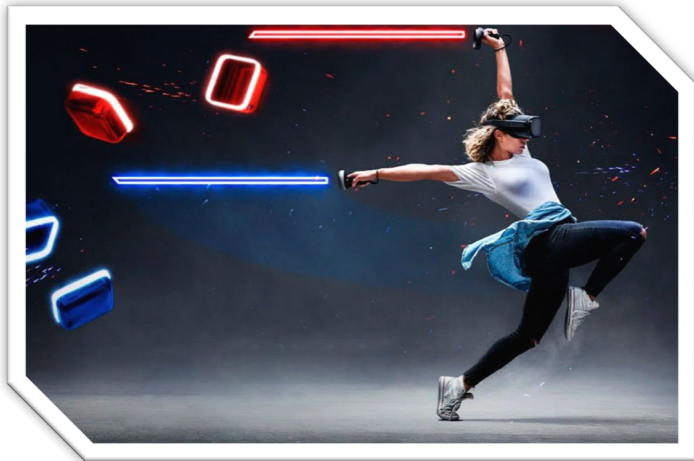
# PERFORMANCE STUDY

## Results

| Identification Measure                      | Macro Average Accuracy | Standard Deviation |
|---------------------------------------------|------------------------|--------------------|
| Cross-session Identification                | 0.93                   | 0.06               |
| Cross-person Identification                 | 0.87                   | 0.09               |
| Cross-person Identification with refinement | 0.91                   | 0.07               |

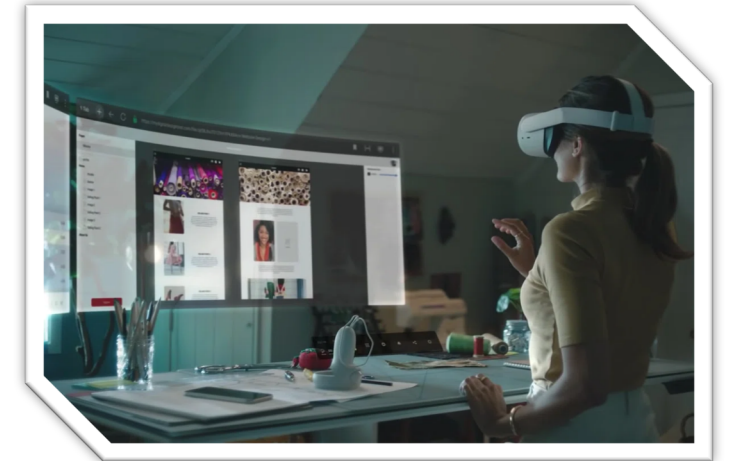


# SCENARIOS



*Beat Saber (by Beat Games)*

- The main success of VR applications so far has been in **Gaming** that engages the whole body.
- The most recent **productivity** applications that attempt to replicate office environments **focus on content consumption** rather than content creation.
- This is a direct result of the **inadequate** interaction modality that is mid-air input, often retrofit to an interface that was designed for keyboard and mice.



*Facebook Infinite Spaces (by Facebook)*

# APPLICATION PROTOTYPES

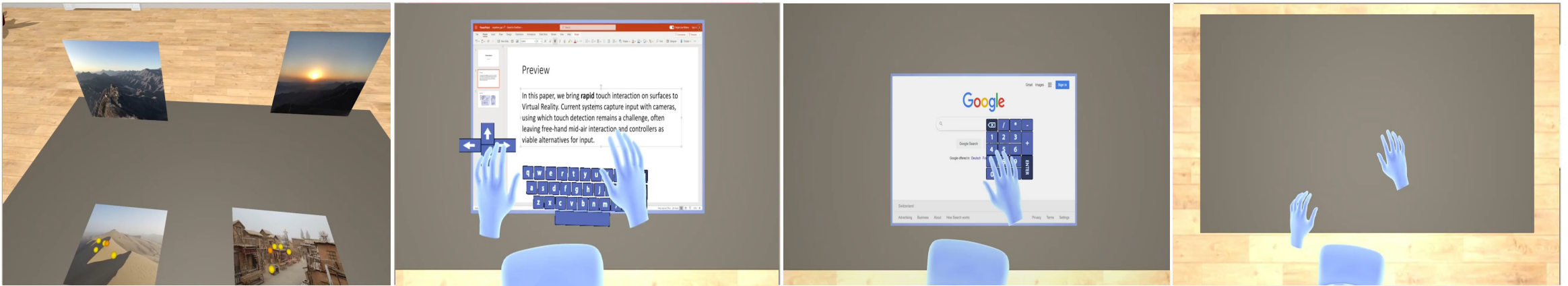


Figure 11: Our demo applications showcase TapID's integration with VR scenes and hand-based interaction using different surface widgets: (a) dragging and resizing photos including a per-finger command input widget below the left hand, (b) editing rich-text documents with keyboard input including text cursor navigation, (c) a numeric keypad for inputting numbers and operations, and (d) virtual piano keys for precise input.

# LIMITATIONS

“

*We have not tuned or evaluated our method on its capability to identify multi-touch events...*

“

“

*Assumption of the use of a table or the user's own body as a passive input surface*

“

“

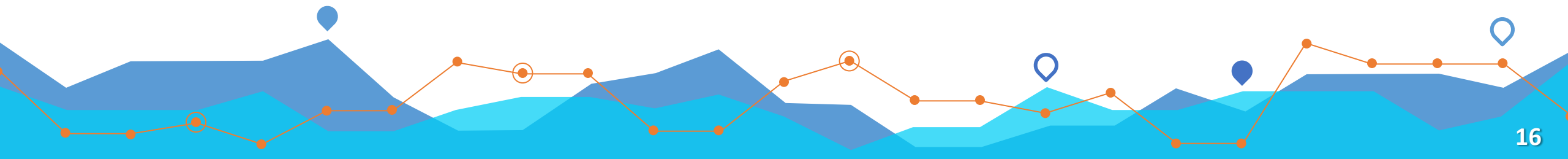
*We did not vary or control tapping strength*

“

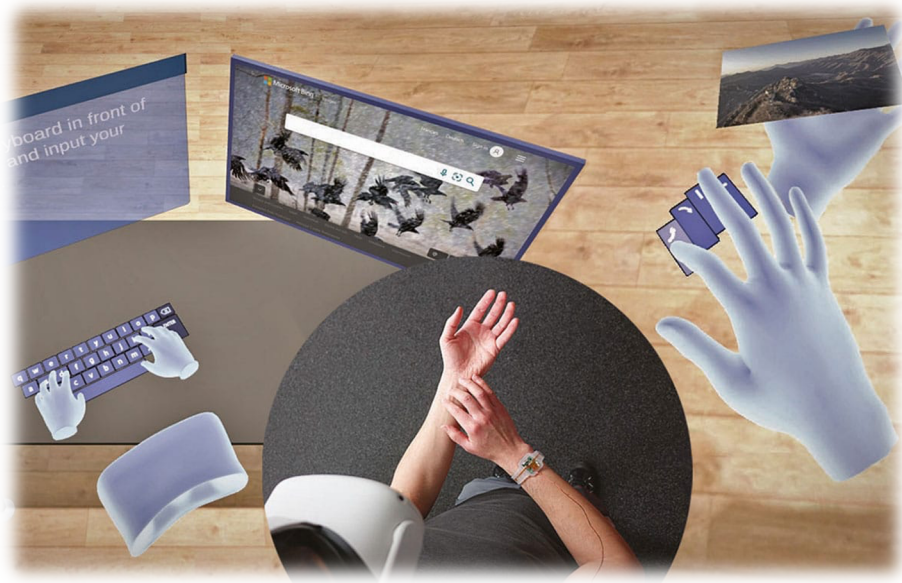
“

*We have tested and evaluated our system in a stationary and seated setting [...]*

“



# CONCLUSION



- Although TapID can be improved, it produced **convincing accuracies** for tap detection and finger identification in a cross-session setting.
- The project is a viable complement for existing VR systems that allows the familiar input modality that is touch input to transfer to **spatial VR systems**, potentially even during everyday scenarios.



## OTHER RELEVANT PAPERS

Sujin Jang, Wolfgang Stuerzlinger, Satyajit Ambike, and Karthik Ramani. 2017. Modeling cumulative arm fatigue in mid-air interaction based on perceived exertion and kinetics of arm motion. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 3328–3339.

Lung-Pan Cheng, Eyal Ofek, Christian Holz, Hrvoje Benko, and Andrew D Wilson. 2017. [Sparse haptic proxy: Touch feedback in virtual environments using a general passive prop](#). In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 3718–3728.

Pascal Knierim, Valentin Schwind, Anna Maria Feit, Florian Nieuwenhuizen, and Niels Henze. 2018. [Physical keyboards in virtual reality: Analysis of typing performance and effects of avatar hands](#). In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–9.

## USED RESOURCES

M. Meier, P. Streli, A. Fender and C. Holz, "TapID: Rapid Touch Interaction in Virtual Reality using Wearable Sensing," *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*, 2021, pp. 519-528, doi: 10.1109/VR50410.2021.00076.

From Google Images:

[https://miro.medium.com/max/640/1\\*oNnQ52r90yp6fVn-pQ5QNQ.png](https://miro.medium.com/max/640/1*oNnQ52r90yp6fVn-pQ5QNQ.png)

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<https://www.itsnicethat.com/news/facebook-infinite-office-digital-210920>

<https://www.researchgate.net/profile/Marco-Speicher-2/publication/320313847/figure/fig1/AS:705891993137154@1545309174716/Bi-manual-mid-air-interaction-in-Virtual-Reality.jpg>

[https://www.kindpng.com/picc/m/31-316503\\_virtual-reality-png-transparent-virtual-reality-png-png.png](https://www.kindpng.com/picc/m/31-316503_virtual-reality-png-transparent-virtual-reality-png-png.png)

